

Linear Models Lecture 10: Asymptotics

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Big Picture

- We began with a model in which $Y = \mathbf{x}'\beta + e$
- OLS finds the projection of $\mathbf{y}$ in the vector space spanned by $\mathbf{x}$:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{P}\mathbf{y}$$

$$\hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\beta} = \mathbf{y} - \mathbf{P}\mathbf{y} = (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{M}\mathbf{y}$$

- We showed that the projection coefficient $\hat{\beta}$ is an unbiased estimator of β .
- With the additional assumption of normality, we derived t and F distributions (Lecture 7).

In Lecture 9, we used asymptotic arguments **informally**: Taylor expansion of the score, LLN for the Hessian, CLT for the score. Today we develop these tools rigorously.

Standards for Inference

- Unbiased estimators ($E[\hat{\theta}] = \theta$) are “correct on average” even in small samples.
 - Unbiasedness tells us that we are not better off with a different guess (we would lose money in expectation).
 - A biased estimator is biased even with infinite data.
- Unbiasedness is not always what we want.
 - Unbiasedness doesn't rule out pathological behavior: the “first impression” estimator is unbiased.
 - Many popular estimators are biased: Probit MLE (Lecture 9), 2SLS, ridge regression.
- A biased estimator may be a good approximation if it gets **closer to the truth with more data**.
- What is a “good” approximation?

Why Consistency Matters: A Concrete Example

U.S. hourly wages: $\mu = 24$, $\sigma^2 = 430$.

“First Impression” estimator

$\hat{\mu} = X_1$ (use only the first observation)

- Unbiased: $\mathbb{E}[X_1] = \mu = 24$
- $\text{Var}(\hat{\mu}) = 430$ for *any* n
- 95% interval: 24 ± 40.7

Sample mean $\bar{X}_n$

- Unbiased: $\mathbb{E}[\bar{X}_n] = \mu = 24$
- $\text{Var}(\bar{X}_n) = 430/n \rightarrow 0$
- 95% interval at $n = 100$: 24 ± 4.1

Both estimators are unbiased. Only $\bar{X}_n$ is **consistent**: its distribution collapses onto μ .

It has been customary to assume (somewhat loosely) that when a quantity is calculated from a random sample to estimate a parameter of a hypothetical frequency distribution, the accuracy of the determination will increase without limit as the number in the sample increases. A *consistent statistic* is a function of the observations actually possessing this property.

- Hotelling 1930

Consistency

- Consistent estimators are “more likely correct with more data”.
- The analog to expectations in large samples is called the probability limit: plim .
- Claim of Consistency: $\text{plim} \hat{\beta} = \beta$, or $\hat{\beta}_j$ is *consistent* for β_j

Convergence of Random Variables

Consider the sample mean $\bar{X}_n$ as a sequence indexed by sample size n . Its distribution changes with n —how do we describe its limit?

There is a hierarchy of convergence concepts for random variables:

$$X_n \xrightarrow{\text{a.s.}} X \quad \Rightarrow \quad X_n \xrightarrow{P} X \quad \Rightarrow \quad X_n \xrightarrow{d} X$$

- **Almost sure (a.s.) convergence** is the strongest form—pathwise convergence except on a probability-zero set. We won't need it operationally.
- **Convergence in probability** ($\xrightarrow{P}$): for all $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$.
 - This is the concept behind **consistency** (WLLN, plim, CMT).
- **Convergence in distribution** ($\xrightarrow{d}$): $F_{X_n}(t) \rightarrow F_X(t)$ at all continuity points.
 - This is the concept behind the **CLT** and asymptotic normality.

Today we develop each in turn: $\xrightarrow{P}$ (next), then $\xrightarrow{d}$ (CLT section).

Convergence in Probability

A sequence of random variables X_n converges *in probability* to c if, for all $\delta > 0$,

$$\lim_{n \rightarrow \infty} \Pr(|X_n - c| \leq \delta) = 1$$

This is written as $X_n \xrightarrow[p]{} c$ or $\text{plim} X_n = c$

We can also say

$$\lim_{n \rightarrow \infty} \Pr(|X_n - c| > \delta) = 0$$

That is, the random variable concentrates about a point in a certain interval.

Unpacking the Definition

$$\lim_{n \rightarrow \infty} \Pr\left(|X_n - \underbrace{c}_{\text{limit value}}| \leq \underbrace{\delta}_{\text{tolerance (fixed first)}}\right) = \underbrace{1}_{\text{4th: in the limit, probability} \rightarrow 1}$$

3rd: take $n \rightarrow \infty$ (after δ is fixed) 2nd: for each n , how often X_n lands within δ of c

Order of operations.

- 1 **Pick** a tolerance $\delta > 0$ (any positive number, however small).
- 2 **Compute** $\Pr(|X_n - c| \leq \delta)$ at each sample size n — a sequence of numbers in $[0, 1]$.
- 3 **Take** the limit of that sequence as $n \rightarrow \infty$.
- 4 **Require** the limit to equal 1, *for every* δ you could have picked.

Not a claim that X_n equals c , or even that $\Pr(|X_n - c| \leq \delta) = 1$ at any finite n .

Plim example

Setup. Z_n takes two values:

$$Z_n = \begin{cases} 0 & \text{w.p. } 1 - p_n \\ a_n & \text{w.p. } p_n \end{cases}$$

When does $Z_n \xrightarrow{P} 0$? Fix any $\delta > 0$ and track $\Pr(|Z_n| \leq \delta)$.

Case 1: the spike shrinks, $a_n \rightarrow 0$.

Eventually $a_n < \delta$, so *both* values of Z_n lie within δ : $\Pr(|Z_n| \leq \delta) = 1$.

Case 2: the spike becomes rare, $p_n \rightarrow 0$.

The mass at 0 alone already covers δ : $\Pr(|Z_n| \leq \delta) \geq 1 - p_n \rightarrow 1$.

Payoff. $E[Z_n] = p_n a_n$ can diverge (let $a_n = n$, $p_n = 1/\sqrt{n}$) while $Z_n \xrightarrow{P} 0$. Probability limits and expectations are not the same object.

Consistency in Practice: OLS with Increasing n

DGP: $Y_i = 2 + 3X_i + \varepsilon_i$, $X_i \sim N(0,1)$, $\varepsilon_i \sim \text{Exp}(1) - 1$ (skewed, non-normal errors).

```
set.seed(42); par(mfrow = c(2, 2))
for (n in c(20, 50, 200, 2000)) {
  betas <- replicate(5000, {
    x <- rnorm(n); y <- 2 + 3*x + (rexp(n)-1)
    coef(lm(y ~ x))[2]
  })
  hist(betas, breaks = 40, prob = TRUE,
       main = paste("n =", n), xlim = c(2, 4))
  abline(v = 3, col = "blue", lwd = 2)
}
```

Errors are skewed and non-normal, yet $\hat{\beta}_1$ concentrates around $\beta_1 = 3$ as n grows.

The Continuous Mapping Theorem

Unlike expectations, plim survives continuous functions.

CMT. If $x_n \xrightarrow{P} x$ and g is continuous at x , then $\text{plim } g(x_n) = g(\text{plim } x_n)$.

If $\text{plim } x_n = c$ and $\text{plim } y_n = d$:

$$\text{plim}(x_n y_n) = cd, \quad \text{plim}(x_n / y_n) = c/d, \quad \text{plim}(x_n + y_n) = c + d.$$

CMT: Why Division Works

The t -statistic divides one random quantity by another. Why is this valid?

- WLLN: $\bar{X}_n \xrightarrow{P} \mu$ and $s_n^2 \xrightarrow{P} \sigma^2$
- CMT (continuous function $g(a, b) = a/\sqrt{b}$):

$$\frac{\bar{X}_n}{s_n} \xrightarrow{P} \frac{\mu}{\sigma}$$

- More generally, any smooth function of consistent estimators is itself consistent.

CMT is what licenses dividing one consistent estimator by another, e.g. $\bar{X}_n/s_n$ in a t -statistic.

Weak Law of Large Numbers

Proof plan:

- Markov's Inequality
- Chebyshev's Inequality
- Weak Law of Large Numbers

Markov's Inequality

Proposition

Suppose X is a random variable that takes on non-negative values. Then, for all $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Markov's Inequality

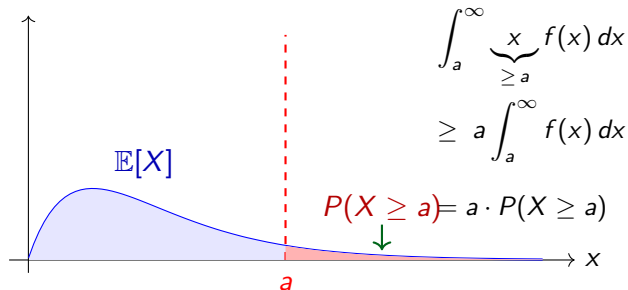
Proof.

For $a > 0$,

$$\begin{aligned} E[X] &= \int_0^{\infty} xf(x)dx \\ &= \int_0^a xf(x)dx + \int_a^{\infty} xf(x)dx \\ E[X] &\geq \int_a^{\infty} xf(x)dx \geq \int_a^{\infty} af(x)dx = aP(X \geq a) \quad (X \geq 0) \\ \frac{E[X]}{a} &\geq P(X \geq a) \end{aligned}$$



Markov's Inequality: Visual Intuition



The full integral $\mathbb{E}[X]$ includes the **tail area**. In the tail, every $x \geq a$, so replacing x with a only makes the integral smaller. Hence $\mathbb{E}[X] \geq a \cdot P(X \geq a)$.

Chebyshev's Inequality

Proposition

If X is a random variable with mean μ and variance σ^2 , then, for any value $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

You will use Chebyshev directly on HW4 Q1(a) (consistency of $\bar{X}_n$) and conditionally on HW4 Q7 (controlling the AIPW remainder).

Chebyshev's Inequality

Proof.

Define $Y = (X - \mu)^2$ where $\mu = E[X]$. Then $Y \geq 0$.

Set $a = k^2$ in Markov's inequality:

$$\begin{aligned} P(Y \geq k^2) &\leq \frac{E[Y]}{k^2} \\ P((X - \mu)^2 \geq k^2) &\leq \frac{E[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2} \\ P(|X - \mu| \geq k) &\leq \frac{\sigma^2}{k^2} \end{aligned}$$



Example of Markov and Chebyshev

Suppose that we know that the number of items produced in a factory is a random variable with mean 500.

- How likely are we to see a production of 1000?
- If the variance is 50, how likely is the production to be between 400 and 600?

Solution

- How likely are we to see a production of 1000?

Markov's Inequality:

$$\mathbb{P}(X \geq 1000) \leq \frac{\mathbb{E}[X]}{1000} = \frac{500}{1000} = 0.5$$

- How likely is production between 400 and 600, given variance = 50?

Chebyshev's Inequality:

$$\mathbb{P}(|X - 500| \geq 100) \leq \frac{50}{100^2} = \frac{1}{200}$$

$$\mathbb{P}(400 \leq X \leq 600) = \mathbb{P}(|X - 500| < 100) \geq 1 - \frac{1}{200} = \frac{199}{200}$$

Chebyshev Sample Size Calculation

- U.S. hourly wages are distributed $\mu = 24$ and $\sigma^2 = 430$,
- How large does n have to be so that $\bar{X}_n$ is within \$1 of the true mean with probability 99%?

$$\mathbb{P} [|\bar{X}_n - \mu| \geq 1] = \mathbb{P} [|\bar{X}_n - \mathbb{E} [\bar{X}_n]| \geq 1] \leq \text{var} [\bar{X}_n] = 430/n$$
$$\frac{430}{n} \leq .01$$

$$\mathbb{P} [|\bar{X}_n - \mu| \geq 1] \leq .01 \text{ if } n = 43,000$$

Weak Law of Large Numbers

Proposition

Suppose $X_1, X_2, \dots, X_n$ is iid random sample from a distribution with expectation μ and $\text{Var}(X_i) = \sigma^2$. Then, for all $\varepsilon > 0$,

$$P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \mu \right| \geq \varepsilon \right\} \rightarrow 0 \text{ as } n \rightarrow \infty$$

or

$$\bar{X}_n \xrightarrow{p} \mathbb{E}[X]$$

Weak Law of Large Numbers

Proof.

$$\frac{\mathbb{E}[X_1 + X_2 + \cdots + X_n]}{n} = \frac{\sum_{i=1}^n \mathbb{E}[X_i]}{n} = \mu$$

$$\mathbb{E} \left[\left(\frac{\sum_{i=1}^n X_i - n\mu}{n} \right)^2 \right] = \frac{\text{Var}(X_1 + X_2 + \cdots + X_n)}{n^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2} = \frac{\sigma^2}{n}$$

$$\mathbb{P} \left(\left| \frac{X_1 + X_2 + \cdots + X_n}{n} - \mu \right| \geq \varepsilon \right) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$



WLLN interpretation

- The sample mean approaches the population mean as $n \rightarrow \infty$
- General version only assumes $\mathbb{E}[X] < \infty$
- This notion that a sequence of the same sample quantity approaches a constant is called consistency.

Any sample moment converges to its population counterpart: $\frac{1}{n} \sum_{i=1}^n X_i X'_i \xrightarrow{P} \mathbb{E}[X_i X'_i]$, $\frac{1}{n} \sum_{i=1}^n X_i Y_i \xrightarrow{P} \mathbb{E}[X_i Y_i]$. This is the foundation for OLS consistency (Lecture 11).

WLLN Counterexample

- WLLN needs to have independent observations:
- For example, assume $X_i = Z + U_i$, like in the random effects model.
- Suppose Z and U_i are random and independent, $\mathbb{E}(U_i) = 0$.
- $\bar{X}_n = Z + \bar{U}_n \xrightarrow[p]{} Z$ not a constant.

Big O and Little o : The Idea

When we take limits, we often need to say “this term is negligible compared to that one.” Asymptotic notation gives us a precise language for this.

- Consider $f(n) = 3n^2 + 5n + 7$ as $n \rightarrow \infty$.
- For large n , the $3n^2$ term dominates: at $n = 100$, we have $30,000 + 500 + 7$.
- We want to say: “ $f(n)$ grows like n^2 ” and “the $5n + 7$ part is *negligible* relative to n^2 .”

Two notions of “negligible”:

- **Little o** = “*vanishes* relative to”: a_n shrinks *faster* than b_n
- **Big O** = “*bounded* relative to”: a_n grows *no faster* than b_n

Analogy: if b_n is a speed limit, then $a_n = o(b_n)$ means a_n is slowing to a stop, while $a_n = O(b_n)$ means a_n is obeying the speed limit (but may still be driving).

Formal Definitions: O and o

Definition

Let $\{a_n\}$ and $\{b_n\}$ be sequences with $b_n > 0$.

■ **Little o :** $a_n = o(b_n)$ means $\frac{a_n}{b_n} \rightarrow 0$ as $n \rightarrow \infty$.

■ **Big O :** $a_n = O(b_n)$ means $\left| \frac{a_n}{b_n} \right| \leq C$ for some constant C and all n large enough.

In words:

■ $a_n = o(b_n)$: “ a_n vanishes compared to b_n ” (the ratio goes to zero)

■ $a_n = O(b_n)$: “ a_n is at most on the order of b_n ” (the ratio stays bounded)

Note: little o is a *stronger* statement than big O . If $a_n = o(b_n)$ then $a_n = O(b_n)$, but not vice versa.

Examples: O and o with Sequences

Statement	Check	Why
$5n = o(n^2)$	$5n/n^2 = 5/n \rightarrow 0 \checkmark$	n grows slower than n^2
$5n = O(n^2)$	$5n/n^2 = 5/n \leq 5 \checkmark$	also bounded (weaker)
$3n^2 = O(n^2)$	$3n^2/n^2 = 3 \checkmark$	bounded but doesn't vanish
$3n^2 = o(n^2)?$	$3n^2/n^2 = 3 \not\rightarrow 0 \times$	same order, not negligible
$1/n = o(1)$	$\frac{1/n}{1} \rightarrow 0 \checkmark$	$1/n$ vanishes
$1/n^2 = o(1/n)$	$\frac{1/n^2}{1/n} = 1/n \rightarrow 0 \checkmark$	$1/n^2$ vanishes faster

Taylor Expansion: A Review

For a smooth function f and a point a , the Taylor series expresses $f(x)$ near a using the derivatives of f at a :

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k = f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 + \cdots$$

Truncating. Keep the first $K+1$ terms; collect the rest as a remainder:

$$f(x) = \underbrace{\sum_{k=0}^K \frac{f^{(k)}(a)}{k!} (x-a)^k}_{\text{polynomial approximation}} + \underbrace{R_K(x)}_{\text{remainder, shrinks as } x \rightarrow a}$$

Example. $f(x) = \ln(1+x)$ at $a = 0$, with $f'(0) = 1$, $f''(0) = -1$, $f'''(0) = 2, \dots$:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

Application: Taylor Remainder in O Notation

Consider the Taylor expansion of $\ln(1+x)$ around $x=0$:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

We can truncate and describe what we dropped:

$$\ln(1+x) = x + O(x^2) \quad (\text{first-order: remainder} \leq Cx^2)$$

$$\ln(1+x) = x - \frac{x^2}{2} + O(x^3) \quad (\text{second-order: remainder} \leq Cx^3)$$

With $x = t/\sqrt{n}$:

- $O(x^2) = O(1/n)$
- $O(x^3) = O(n^{-3/2})$

The CLT proof drops the Taylor remainder in the MGF expansion using exactly this scaling.

Big O_p and Little o_p : The Stochastic Versions

For *random* variables, we add a subscript p (“in probability”):

Definition

- $X_n = o_p(a_n)$: $X_n/a_n \xrightarrow{p} 0$ (“vanishes in probability”)
- $X_n = O_p(a_n)$: X_n/a_n stays bounded in probability

Translations to what we just proved:

- $X_n \xrightarrow{p} c$ is the same as writing $X_n - c = o_p(1)$
- WLLN says $\bar{X}_n - \mu = o_p(1)$, and more precisely $\bar{X}_n - \mu = O_p(n^{-1/2})$
- CLT (coming next) says $\sqrt{n}(\bar{X}_n - \mu) = O_p(1)$

$o_p(1)$ in a proof denotes a term that vanishes in probability. *HW4 Q7: show the AIPW remainder is $o_p(n^{-1/2})$.*

Slutsky's Lemma: Combining $\xrightarrow{p}$ and $\xrightarrow{d}$

Proposition (Slutsky's Lemma)

Suppose $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ (a constant). Then

- $X_n + Y_n \xrightarrow{d} X + c$
- $X_n Y_n \xrightarrow{d} c X$
- $X_n / Y_n \xrightarrow{d} X / c$ (if $c \neq 0$)

A random limit for Y_n is not enough — it must settle on a constant.

Special case. If $R_n = o_p(1)$, then $X_n + R_n \xrightarrow{d} X$ — an $o_p(1)$ remainder cannot affect a limiting distribution.

The recipe: $\sqrt{n}(\hat{\theta} - \theta) = \underbrace{(\text{CLT term})}_{\xrightarrow{d} N} + \underbrace{R_n}_{o_p(1)}$. Slutsky discards R_n .

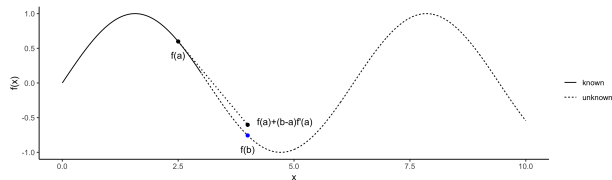
Used on HW4 Q1(b), Q4(e), Q7–Q8.

Taylor Approximation

We know $f(a)$ and its derivatives at a , and want to approximate $f(b)$ for b near a . Build the approximation one derivative at a time:

$$f(b) = f(a) + f'(a)(b - a) + \frac{1}{2}f''(a)(b - a)^2 + O((b - a)^3).$$

The $O((b - a)^3)$ remainder shrinks fast: if $b - a = t/\sqrt{n}$, then $(b - a)^2 = t^2/n$ and $(b - a)^3 = O(n^{-3/2})$ — negligible compared to the first two terms.



The delta method uses the first-order expansion; the CLT proof uses the second-order expansion of the MGF.

Three Tools the CLT Proof Will Need

1. Taylor expansion (2nd order): $f(b) = f(a) + f'(a)(b-a) + \frac{1}{2}f''(a)(b-a)^2 + O((b-a)^3)$.
For $b-a = t/\sqrt{n}$, the O term is $O(n^{-3/2})$.

2. Exponential limit: $(1 + x/n)^n \rightarrow e^x$. (Take logs, Taylor-expand $\ln(1 + x/n)$, multiply by n .)

3. Moment generating function: $M_X(t) = \mathbb{E}[e^{tX}]$. Three properties we use:

- $M_X^{(k)}(0) = \mathbb{E}[X^k]$ (Taylor expansion in t generates moments).
- Independence: $M_{X_1+\dots+X_n}(t) = [M_X(t)]^n$ for iid sums.
- Continuity: $M_{X_n}(t) \rightarrow M_X(t)$ near 0 $\Rightarrow X_n \xrightarrow{d} X$.

Target: $Z \sim N(0, 1) \Rightarrow M_Z(t) = e^{t^2/2}$. CLT proof shows the MGF of the standardized sample mean converges to this.

Convergence in Distribution

- We want the sampling distribution of the sample mean $\bar{X}_n$.
- The sampling distribution is a function of the distribution of observations F and the sample size n .
- We will get an *asymptotic approximation* by standardizing $\bar{X}_n$ and taking the limit as $n \rightarrow \infty$

Definition: Convergence in Distribution

- Let X_n be a sequence of random variables, and X a target random variable.
- We say that X_n **converges in distribution** to X if:

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq t) = \mathbb{P}(X \leq t) \quad \text{for all } t \text{ where } F_X(t) \text{ is continuous}$$

- **Implications**

- The cumulative distribution functions converge: $F_{X_n}(t) \rightarrow F_X(t)$
- If the target distribution is a constant, then this is just convergence in probability

- **Notation:** $X_n \xrightarrow{d} X$

Standardizing the Sample Mean

- The WLLN tells us that $\bar{X}_n \xrightarrow{p} \mu$, so $\bar{X}_n \xrightarrow{d} \mu$.
- But the distribution of μ is degenerate, not useful.
- So rescale by the variance: $\text{var}[\bar{X}_n - \mu] = \frac{\sigma^2}{n}$, so $\text{var}[\sqrt{n}(\bar{X}_n - \mu)] = \sigma^2$

$$Z_n = \sqrt{n}(\bar{X}_n - \mu)$$

- As $n \rightarrow \infty$, $\mathbb{E}[Z_n] \rightarrow 0$, $\mathbb{E}[Z_n^2] \rightarrow \sigma^2$, $\mathbb{E}[Z_n^3] \rightarrow 0$, $\mathbb{E}[Z_n^4] \rightarrow 3\sigma^4$
- The Central Limit Theorem states that $Z_n \xrightarrow{d} Z$ where $Z \sim N(0, \sigma^2)$.
- This means that $\bar{X}_n \underset{a}{\sim} N(\mu, \frac{\sigma^2}{n})$.

The Central Limit Theorem

Theorem (Lindeberg–Lévy CLT)

Let $X_1, X_2, \dots$ be iid with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Equivalently: $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$.

Key conditions:

- **iid** — can be relaxed (Lindeberg–Feller allows heterogeneity)
- **Finite variance** — essential; without it, convergence may fail or be to a non-normal limit

The limit distribution is $N(0, \sigma^2)$ regardless of the distribution of X_i — Bernoulli, exponential, or uniform data all yield an approximately normal sample mean for large n . (HW4 Q4(e) applies this to an influence-function average $\psi(Z_i)$.)

Sketch of the CLT Proof

Goal. Show $M_{Z_n}(t) \rightarrow e^{t^2/2}$ for $Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$. The continuity theorem then gives $Z_n \xrightarrow{d} N(0, 1)$.

Standardize $W_i = (X_i - \mu)/\sigma$ so $\mathbb{E}[W_i] = 0$, $\mathbb{E}[W_i^2] = 1$, and $Z_n = \frac{1}{\sqrt{n}} \sum_i W_i$. By independence,

$$M_{Z_n}(t) = [M_W(t/\sqrt{n})]^n.$$

Taylor-expand M_W around 0, using $M_W(0) = 1$, $M'_W(0) = \mathbb{E}[W] = 0$, $M''_W(0) = \mathbb{E}[W^2] = 1$:

$$M_W(s) = 1 + \frac{1}{2}s^2 + O(s^3) \implies M_W(t/\sqrt{n}) = 1 + \frac{t^2}{2n} + O(n^{-3/2}).$$

Raise to the n -th power and apply the exponential limit:

$$M_{Z_n}(t) = \left[1 + \frac{t^2/2}{n} + O(n^{-3/2}) \right]^n \longrightarrow e^{t^2/2}. \quad \blacksquare$$

Recipe: Taylor-expand the MGF, raise to n via independence, take the exponential limit.

CLT Sample Size Calculation

- U.S. wages: $\mu = 24$, $\sigma^2 = 430$. How large must n be so $\bar{X}_n$ is within \$1 of μ with 99% probability?

$$\mathbb{P}[|\bar{X}_n - \mu| \geq 1] \underset{a}{\sim} \mathbb{P}[|Z| \geq \sqrt{n/430}] = .01$$

$$\sqrt{n/430} = 2.576 \quad \Rightarrow \quad n = 430 \times 2.576^2 \geq 2,862$$

Compare with Chebyshev: $n = 43,000$. The CLT is **15× more efficient** because it uses the shape of the distribution, not just the variance.

CLT: R Simulation

```
# CLT: sample means of Exponential(1) converge to Normal
set.seed(42)
par(mfrow = c(2, 2))
for (n in c(5, 30, 100, 1000)) {
  xbar <- replicate(10000, mean(rexp(n, rate = 1)))
  hist(xbar, breaks = 40, prob = TRUE,
       main = paste("n =", n),
       xlab = expression(bar(X)[n]))
  curve(dnorm(x, 1, 1/sqrt(n)), add = TRUE,
        col = "red", lwd = 2)
}
```

The red curve is the normal approximation $N(\mu, \sigma^2/n)$. Even for **skewed** distributions (Exponential), the normal approximation is excellent by $n = 30$.

When Do Asymptotics Kick In?

Data shape	Rule of thumb	Example
Symmetric, light tails	$n \geq 30$	Heights, test scores
Moderate skew	$n \geq 100$	Wages, income
Heavy tails / outliers	$n \geq 500+$	Financial returns, conflict
Rare binary outcome	Events ≥ 10 /covariate	Coups, wars

The CLT convergence rate depends on the shape of the underlying distribution; the cutoffs above are rules of thumb, not theorems.

Multivariate CLT

The CLT generalizes to vectors. If $\mathbf{W}_1, \dots, \mathbf{W}_n$ are iid k -vectors with $\mathbb{E}[\mathbf{W}_i] = \boldsymbol{\mu}$ and $\text{Var}(\mathbf{W}_i) = \boldsymbol{\Sigma}$, then:

$$\sqrt{n}(\bar{\mathbf{W}}_n - \boldsymbol{\mu}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma})$$

Applications:

- **OLS** (Lecture 11): $\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i \xrightarrow{d} N(\mathbf{0}, \sigma^2 Q_{XX})$
- **Probit MLE** (Lecture 9): $\frac{1}{\sqrt{n}} S_n(\beta_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\beta_0))$
- **GMM** (coming later): $\frac{1}{\sqrt{n}} \sum_{i=1}^n g(W_i, \theta_0) \xrightarrow{d} N(\mathbf{0}, \Omega)$

In each case, a sum of mean-zero random vectors converges to a multivariate normal. The covariance matrix differs, but the **structure** is identical.

What We Developed Today

Tool	Statement	Application
WLLN	$\bar{X}_n \xrightarrow{P} \mu$	Consistency
CMT	$\text{plim } g(X_n) = g(\text{plim } X_n)$	Functions of estimators
O_p, o_p	Rate of convergence	Tracking remainders
CLT	$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$	Asymptotic normality

The CLT proof brought together:

- 1 Taylor expansion of the MGF (second-order approximation)
- 2 $O(\cdot)$ notation to track the remainder
- 3 The exponential limit $(1 + x/n)^n \rightarrow e^x$
- 4 MGF continuity theorem to conclude convergence in distribution

Looking Ahead

Lecture 11 introduces the **Delta Method** and applies all of today's tools to OLS:

- **Delta Method:** Taylor expansion + CLT $\Rightarrow$ distribution of $h(\hat{\beta})$
- **Consistency:** WLLN $\Rightarrow \frac{1}{n}\mathbf{X}'\mathbf{X} \xrightarrow{P} Q_{XX}$, CMT $\Rightarrow \hat{\beta} \xrightarrow{P} \beta$
- **Asymptotic normality:** Multivariate CLT $\Rightarrow \sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V_\beta)$
- **Wald tests:** Delta method $\Rightarrow$ test nonlinear hypotheses about β

These tools also justify what we did informally in Lecture 9 (Probit):

- Score CLT: $\frac{1}{\sqrt{n}}S_n(\beta_0) \xrightarrow{d} N(0, \mathcal{I}(\beta_0))$ (multivariate CLT)
- Hessian convergence: $\frac{1}{n}H_n(\beta_0) \xrightarrow{P} -\mathcal{I}(\beta_0)$ (WLLN)
- Combining via Slutsky: $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, \mathcal{I}^{-1})$